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# Participation Report on the Organisation of the CTF ISTEC Porto 2026

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# Index

Introduction ..... 5
Background and Organisation of the Event ..... 6
About ISTEC Porto ..... 6
Format, platform and links ..... 6
Categories and challenges ..... 7
Organising team ..... 8
Rules and Regulations ..... 11
Preparation Session Bootcamp: 22 October ..... 11
CTF Day: 29 October ..... 12
Accreditation and opening session ..... 12
Competition ..... 12
Coffee break ..... 14
Walkthroughs and prize-giving ..... 15
Results and Statistics ..... 15
Overview ..... 15
Breakdown by category and points ..... 16
Solves per challenge ..... 17
Score and submission distribution ..... 18
Progression and list of teams ..... 19
Final leaderboard ..... 20
Podium, Winners and Recognition ..... 21
1st place: Marrocos ..... 21
2nd place: Ciberpunks ..... 22
3rd place: Gondomar Sport Clube ..... 23
"Rookie" prize ..... 24
Trophies and keepsakes ..... 25
Critical Analysis and Recommendations ..... 26
Conclusion ..... 27



# Table of Figures

Figure 1: Home page of the CTFd platform for the CTF ISTEC Porto 2026. ..... 6
Figure 2: Part of the CTF ISTEC Porto organising team. ..... 8
Figure 3: Organising team of the CTF ISTEC Porto (2). ..... 9
Figure 4: Organising team of the CTF ISTEC Porto (3). ..... 9
Figure 5: Organising team of the CTF ISTEC Porto (4). ..... 10
Figure 6: Competition room with the participating teams during the CTF. ..... 12
Figure 7: Participants solving challenges during the competition. ..... 13
Figure 8: Coffee break offered to the participants. ..... 14
Figure 9: Overview of the platform statistics: 88 users, 26 teams, 43 challenges and 8100 points. ..... 15
Figure 10: Breakdown of challenges by category. ..... 16
Figure 11: Breakdown of points by category. ..... 16
Figure 12: Number of solves per challenge. ..... 17
Figure 13: Percentage of teams that solved each challenge. ..... 17
Figure 14: Distribution of team scores. ..... 18
Figure 15: Percentage of correct and incorrect submissions. ..... 18
Figure 16: Total of correct and incorrect submissions. ..... 18
Figure 17: Score progression of the top 10 teams throughout the competition. ..... 19
Figure 18: List of teams registered on the platform. ..... 19
Figure 19: Final leaderboard (1st to 15th place). ..... 20
Figure 20: Final leaderboard (15th to 22nd place). ..... 20
Figure 21: Winning team, Marrocos (1st place), with the organisation. ..... 21
Figure 22: Ciberpunks team (2nd place), with the organisation. ..... 22
Figure 23: Gondomar Sport Clube team (3rd place), with the organisation. ..... 23
Figure 24: Team awarded the "Rookie" (Revelação) prize, with the organisation. ..... 24
Figure 25: CTF ISTEC Porto trophy, 3D-printed. ..... 25
Figure 26: CTF ISTEC Porto keychain, 3D-printed. ..... 25



# Introduction

This report documents my participation in the organisation of the 1st edition of the CTF ISTEC Porto: Capture the Flag of Cybersecurity, a hands-on, jeopardy-style cybersecurity competition promoted by ISTEC Porto (Instituto Superior de Tecnologias Avançadas do Porto). The event took place over two dates: a preparation session (bootcamp) on 22 October 2026, and the competition itself on 29 October 2026.

In the CTF, 22 teams of 2 to 4 members competed, solving more than 30 challenges across several categories, ranging from beginner-level to areas such as OSINT, CTI, Web, Forensics, Cryptography, Reverse and Programming/Logic. As a member of the organising team, I contributed to the conception, preparation and running of the event, as well as to supporting the competition and the participants.

This report also includes a list of figures that visually accompanies the main moments, venues and results of the event, helping to contextualise each stage I describe.

# Background and Organisation of the Event

## About ISTEC Porto

The CTF was promoted by ISTEC Porto (Instituto Superior de Tecnologias Avançadas do Porto), a higher-education institution in the field of technology. The 1st edition aimed to bring students closer to cybersecurity, to foster problem-solving, critical thinking and teamwork, and it was open to students with no prior experience.

## Format, platform and links

The competition followed the jeopardy format, in which teams solve challenges to find codes ("flags") that award points. As a rule, flags had the format ISTEC{text} and were case-sensitive. The competition ran on a CTFd platform instance, hosted on infrastructure within the European Union.




Figure 1: Home page of the CTFd platform for the CTF ISTEC Porto 2026.



# Categories and challenges

The platform featured 43 challenges spread across eight categories with different difficulty levels. Each challenge had a fixed value of 100, 200, 300 or 400 points, depending on its difficulty, for a total of 8100 points at stake. Some challenges offered hints: some free, some costing points, and certain challenges depended on first solving others.

The distribution of challenges across categories was as follows: Web (18), OSINT (7), Misc (6), Cryptography (4), Forensics (3), Reverse (2), CTI (1), Beginner (1) and Programming/Logic (1).

# Organising team

The event was organised by the lecturer Mário Amorim (coordination) and the technician Nuno Fonseca, with the support of students from the BSc in Computer Engineering. The team included:

- **Mário Amorim:** Lecturer at ISTEC Porto;
- **Nuno Fonseca:** Technician at ISTEC Porto;
- **Vanina Kollen**: Student;
- **Ana Rita Monteiro:** Student;
- **Rafael Batista:** Student;
- **José Samuel Oliveira (Ph0enix):** Student.

Unfortunately, one of the student organisers, Rafael, was unable to attend on the day of the CTF, and another organiser, Vanina, had to leave midway through the event.




**Figure 2: Part of the CTF ISTEC Porto organising team.**



Figure 3: Organising team of the CTF ISTEC Porto (2).



Figure 4: Organising team of the CTF ISTEC Porto (3).



Figure 5: Organising team of the CTF ISTEC Porto (4).



# Rules and Regulations

The event was governed by its own set of regulations, approved for the 1st edition. The main points are:

- **Eligibility:** open to students of any course and year at ISTEC Porto, with no prior cybersecurity experience required.
- **Teams:** participation in teams of 2 to 4 students, each team with at least one laptop.
- **Registration:** free, from 15 to 26 October 2026, using the institutional email address (@my.istec.pt), limited to the room capacity.
- **Scoring:** challenges worth 100 to 500 points; paid hints deducted from the score; ties broken by the team that reached the score first and then by the number of challenges solved.
- **Conduct:** use of the Internet, public documentation, security tools and generative artificial intelligence was allowed; attacking the infrastructure, sharing flags between teams, automating submissions or using multiple accounts was forbidden.
- **Prizes and certificates:** prizes for the top three teams, a "Rookie" (Revelação) prize for the best eligible team, and a certificate of participation for everyone who stayed until the end.

The regulations also included data protection (GDPR) and image-rights rules, with ISTEC Porto acting as the controller of the data collected.

# Preparation Session Bootcamp: 22 October

On 22 October 2026 a preparation session (bootcamp) was held, optional and especially recommended for those who had never taken part in a CTF. In this session we explained to participants how to approach and solve challenges across the various categories, presenting the tools best suited to each one. The goal was to lower the barrier to entry and allow students with no experience to arrive at competition day more confident.

# CTF Day: 29 October

## Accreditation and opening session

The competition took place on 29 October 2026, at the ISTEC Porto premises. Accreditation began at 1:30 p.m., confirming the students' attendance, and the opening session was held at 2:00 p.m.

## Competition

The competition ran from 2:15 p.m. to 5:15 p.m., bringing together 22 teams of 2 to 4 members who competed against each other to solve the challenges. Throughout the afternoon, the organisation supported the teams, answering questions about the platform and the regulations, without helping to solve the challenges.



**Figure 6: Competition room with the participating teams during the CTF.**



Figure 7: Participants solving challenges during the competition.

# Coffee break

During the event a coffee break was offered to the participants, providing a moment to pause and to socialise between the teams and the organisation.



Figure 8: Coffee break offered to the participants.

# Walkthroughs and prize-giving

After the competition closed and the complaints were reviewed, from 5:30 p.m. onwards there were walkthroughs of some of the challenges and the prize-giving to the winning teams.

# Results and Statistics

## Overview

The platform recorded 88 users and 26 teams (of which 22 scored and appear in the final leaderboard), from 265 distinct IP addresses. There were 43 challenges and 8100 points at stake. The most-solved challenge was "Lê as Regras" (22 solves) and the least-solved was Ecos of the Past (2/3) (1 solve).

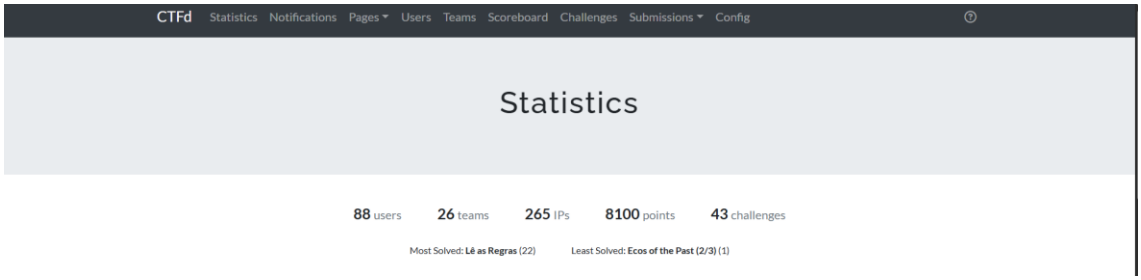



Figure 9: Overview of the platform statistics: 88 users, 26 teams, 43 challenges and 8100 points.

# Breakdown by category and points

The figures below show the distribution of challenges and points by category. The Web category was by far the most represented, with 18 challenges (41.9%) and 3500 points (43.2%), followed by OSINT, with 7 challenges and 1500 points.

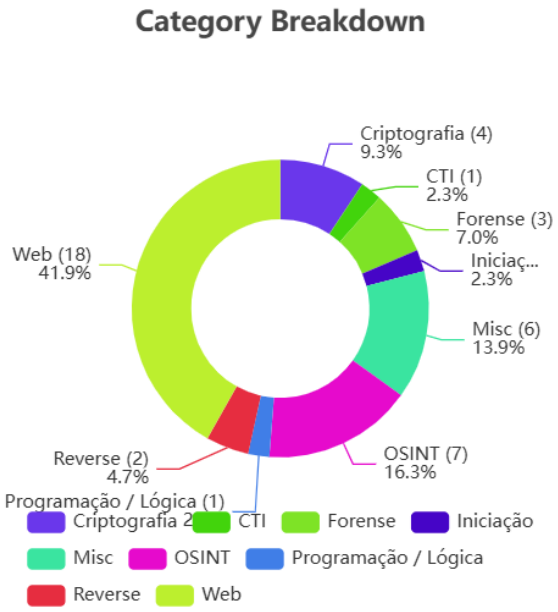



Figure 10: Breakdown of challenges by category.

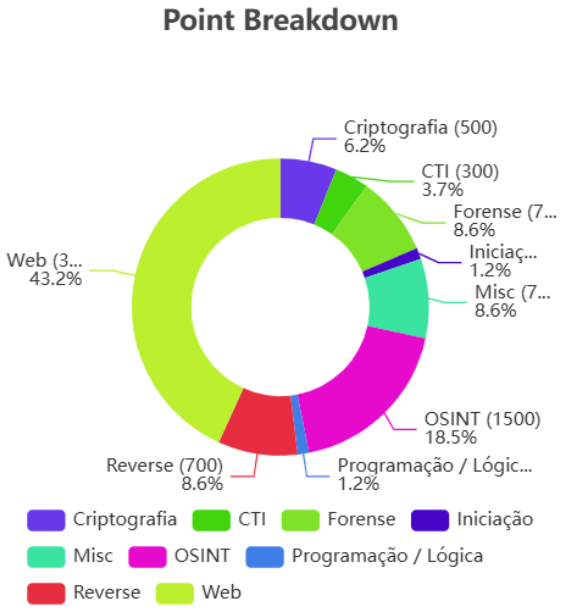



Figure 11: Breakdown of points by category.

# Solves per challenge

Looking at the challenges individually, there is a wide range in the number of solves: some beginner challenges were solved by almost every team, while the hardest ones went unsolved or were solved by only one or two teams.

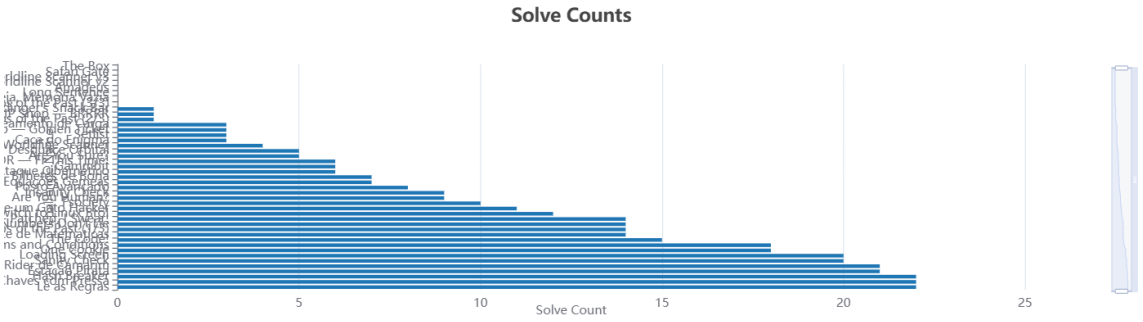



Figure 12: Number of solves per challenge.

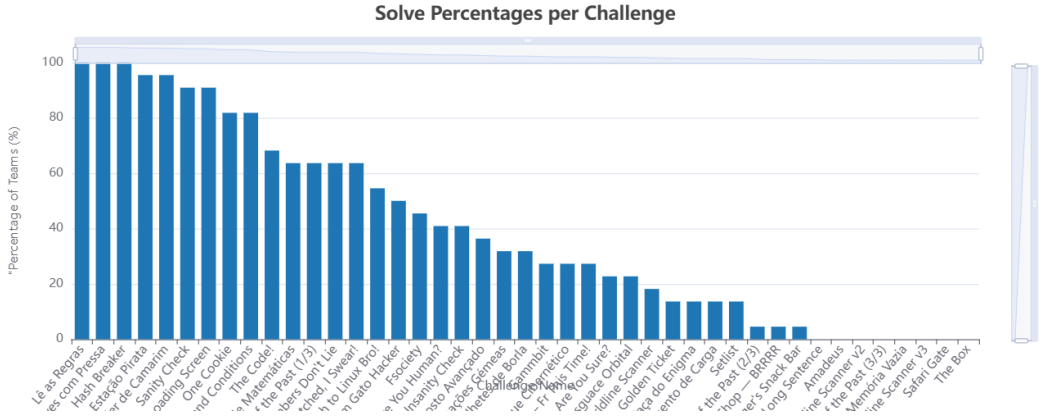



Figure 13: Percentage of teams that solved each challenge.

# Score and submission distribution

Over the course of the competition 2028 flag submissions were made, of which 379 were correct (18.7%) and 1649 were incorrect (81.3%). The distribution of final team scores is fairly spread out, with most teams scoring between 900 and 2000 points.

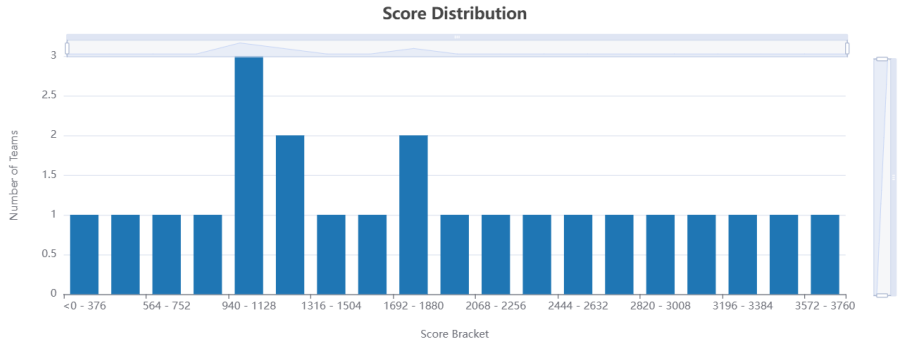



Figure 14: Distribution of team scores.

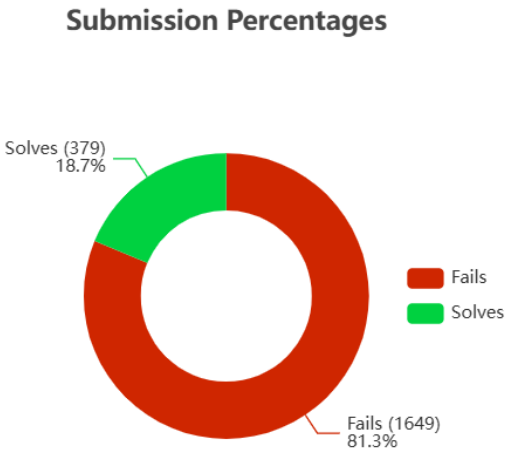



Figure 15: Percentage of correct and incorrect submissions.

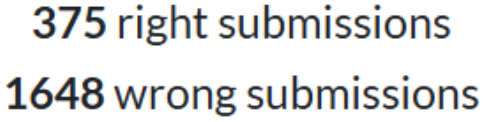



Figure 16: Total of correct and incorrect submissions.

# Progression and list of teams

The score progression of the top 10 teams throughout the afternoon shows a closely-fought competition, with several changes of position right up to the final part of the event.

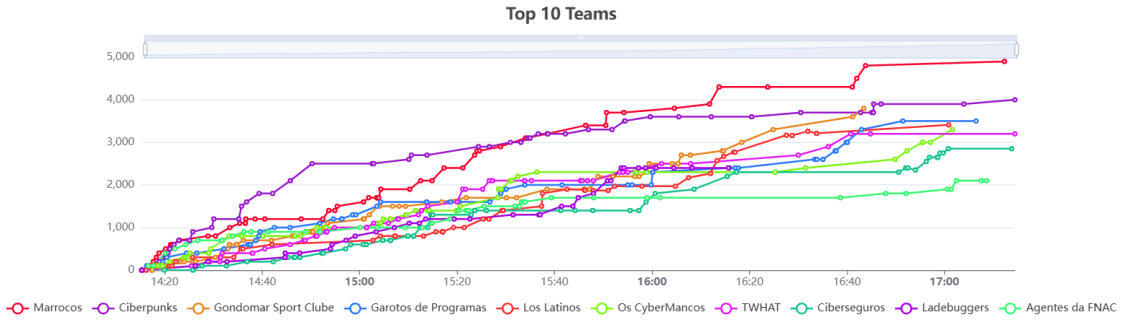



Figure 17: Score progression of the top 10 teams throughout the competition.

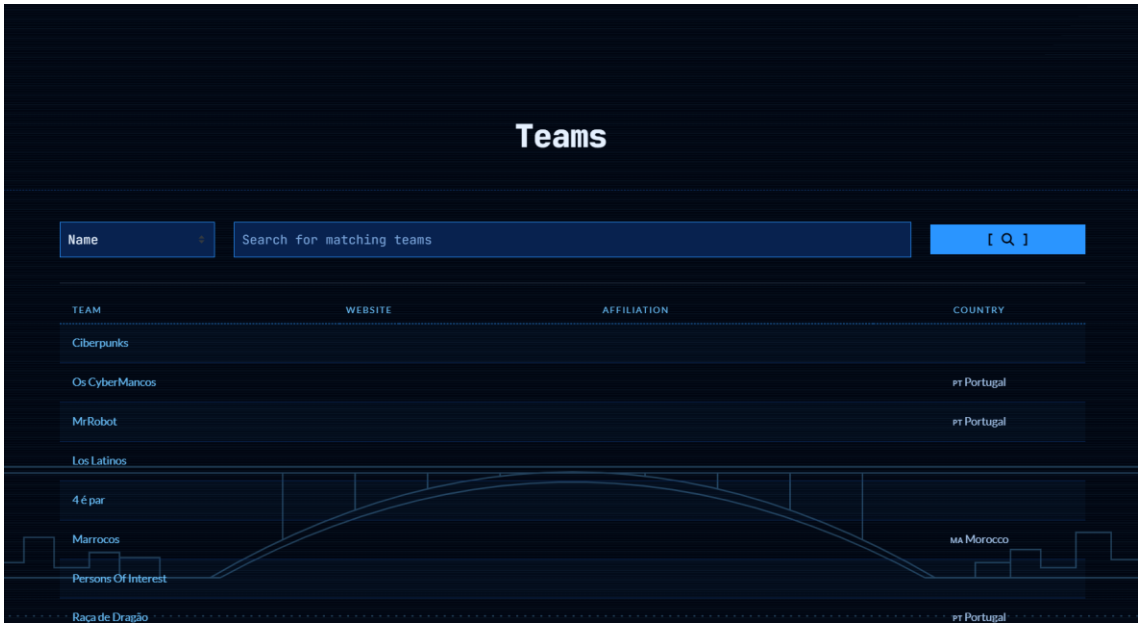



Figure 18: List of teams registered on the platform.

# Final leaderboard

The final leaderboard of the 22 scoring teams was led by the team "Marrocos" (4900 points), followed by "Ciberpunks" (4000) and "Gondomar Sport Clube" (3800). The figures below show the complete table.



| PLACE | TEAM | SCORE |
|---|---|---|
| 1 | Marrocos | 4900 |
| 2 | Ciberpunks | 4000 |
| 3 | Gondomar Sport Clube | 3800 |
| 4 | Garotos de Programas | 3500 |
| 5 | Los Latinos | 3410 |
| 6 | Os CyberMancos | 3300 |
| 7 | TWHAT | 3200 |
| 8 | Ciberseguros | 2850 |
| 9 | Ladebuggers | 2400 |
| 10 | Agentes da FNAC | 2100 |
| 11 | MrRobot | 1750 |
| 12 | Stark-Industries | 1700 |
| 13 | Teletubbies do Bairro | 1600 |
| 14 | Warmongers | 1500 |
| 15 | 4 é par | 1250 |

Figure 19: Final leaderboard (1st to 15th place).



| | | |
|---|---|---|
| 15 | 4 é par | 1250 |
| 16 | Programadores fc | 1200 |
| 17 | Goblin Gigante | 1000 |
| 18 | Persons Of Interest | 1000 |
| 19 | wifi_da_vizinha | 1000 |
| 20 | Raça de Dragão | 845 |
| 21 | Pirin-Pimpim | 630 |
| 22 | FULL BOX 200 | 425 |

Powered by CTFd

Figure 20: Final leaderboard (15th to 22nd place).

# Podium, Winners and Recognition

## 1st place: Marrocos

The winning team, Marrocos, finished in 1st place with 4900 points. It was made up of:

- **Gonçalo Manuel Albuquerque Marinho**;
- **Filipe Calatré**;
- **Bruno Lamela**;
- **Luiz Peneda.**



Figure 21: Winning team, Marrocos (1st place), with the organisation.

# 2nd place: Ciberpunks

Second place was taken by the team Ciberpunks, with 4000 points.



Figure 22: Ciberpunks team (2nd place), with the organisation.

# 3rd place: Gondomar Sport Clube

The third spot on the podium was taken by the team Gondomar Sport Clube, with 3800 points.




Figure 23: Gondomar Sport Clube team (3rd place), with the organisation.

# "Rookie" prize

The "Rookie" prize was awarded to the best-placed eligible team whose members were, individually, first-year students and/or students from courses outside the IT field. The awarded team received an FNAC voucher.



Figure 24: Team awarded the "Rookie" (Revelação) prize, with the organisation.

# Trophies and keepsakes

The trophies for the winning teams and the commemorative keychains handed out were 3D-printed, with a cybersecurity-themed design. Everyone who stayed until the end of the competition also received a certificate of participation.




Figure 25: CTF ISTEC Porto trophy, 3D-printed.




Figure 26: CTF ISTEC Porto keychain, 3D-printed.



# Critical Analysis and Recommendations

Overall, the 1st edition of the CTF ISTEC Porto was very positive. Turnout was strong, the competition ran without any significant incidents, and the atmosphere was one of great enthusiasm and good spirits, fulfilling the goal of bringing students closer to cybersecurity. The preparation session proved important for including participants with no prior experience.

Even so, I leave a few recommendations for future editions:

- Strengthen the balance between categories, since the Web category concentrated a large share of the challenges and points.
- Add more intermediate-difficulty challenges, to reduce the gap between the beginner challenges and the most advanced ones.
- Keep promoting the preparation session (bootcamp) and, if possible, provide supporting materials ahead of the event.
- Maintain and value recognition initiatives, such as the "Rookie" prize, which encourage participation from students of other fields and years.



# Conclusion

Taking part in the organisation of the 1st edition of the CTF ISTEC Porto was a very rewarding experience, both technically and personally. It was fulfilling to see dozens of students competing, learning and having fun with cybersecurity, many of them for the first time.

I hope this first edition is the start of a tradition at ISTEC Porto and that future editions are even bigger, more diverse and more challenging, consolidating cybersecurity as an area of interest and of the future for students.

# Obrigado

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